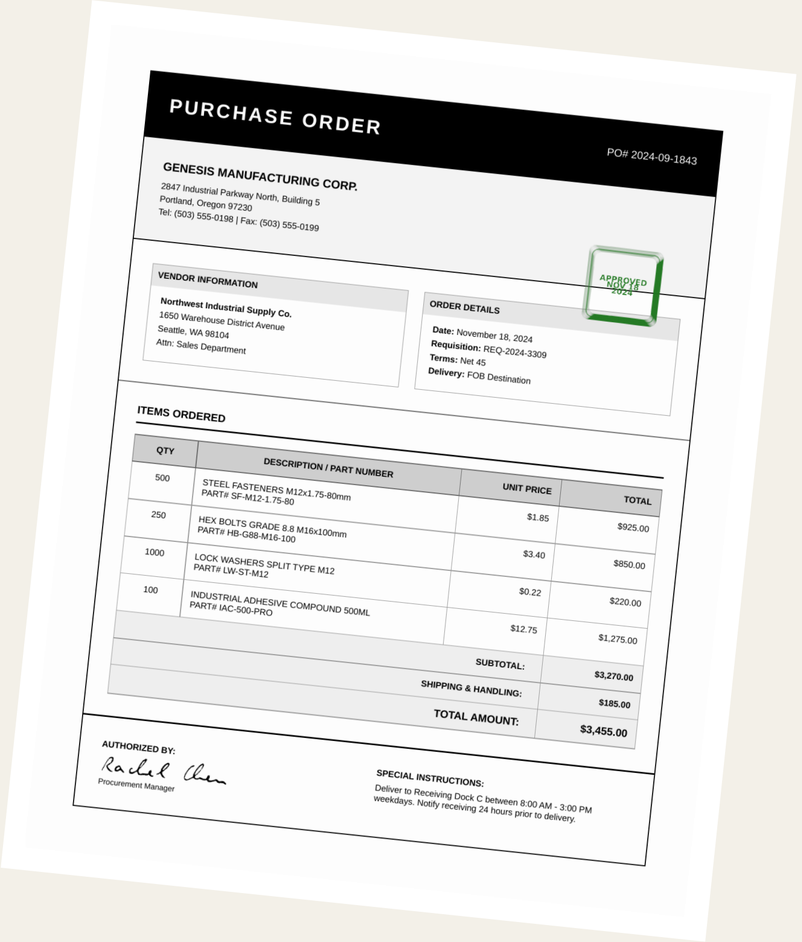
From Handwriting *to Hidden Forgeries*

How reliable is Document AI?

Nauman Riaz

RPTU Kaiserslautern-Landau · DFKI



PURCHASE ORDER

PO# 2024-09-1843

GENESIS MANUFACTURING CORP.
2847 Industrial Parkway North, Building 5
Portland, Oregon 97230
Tel: (503) 555-0198 | Fax: (503) 555-0199

APPROVED

VENDOR INFORMATION
Northwest Industrial Supply Co.
1650 Warehouse District Avenue
Seattle, WA 98104
Attn: Sales Department

ORDER DETAILS
Date: November 18, 2024
Requisition: REQ-2024-3309
Terms: Net 45
Delivery: FOB Destination

ITEMS ORDERED

QTY	DESCRIPTION / PART NUMBER	UNIT PRICE	TOTAL
500	STEEL FASTENERS M12x1.75-80mm PART# SF-M12-1.75-80	$1.85	$925.00
250	HEX BOLTS GRADE 8.8 M16x100mm PART# HB-G88-M16-100	$3.40	$850.00
1000	LOCK WASHERS SPLIT TYPE M12 PART# LW-ST-M12	$0.22	$220.00
100	INDUSTRIAL ADHESIVE COMPOUND 500ML PART# IAC-500-PRO	$12.75	$1,275.00
		SUBTOTAL:	$3,270.00
		SHIPPING & HANDLING:	$185.00
		TOTAL AMOUNT:	$3,455.00

AUTHORIZED BY:
Rachel Chen
Procurement Manager

SPECIAL INSTRUCTIONS:
Deliver to Receiving Dock C between 8:00 AM - 3:00 PM weekdays. Notify receiving 24 hours prior to delivery.

Created by DocDjinn

SYNTHETIC DOCUMENT

Where we *left off.*

DHSD

37 writers

German city and street names



Real dataset samples · not generated handwriting

The model contribution

StylusAI

Text + writer style

+ printed character guide

↓



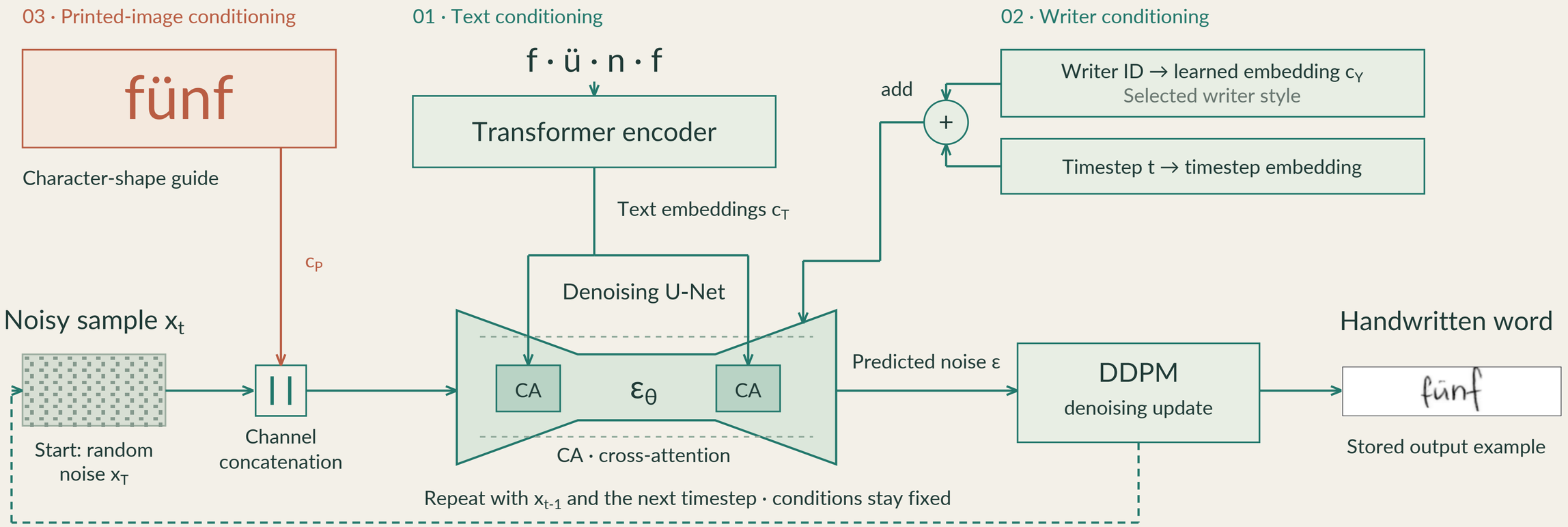
Generated handwriting

THEN, A FUTURE DIRECTION

Document forgery detection →

First colloquium: DHSD + StylusAI; forgery detection as an outlook (PDF p. 26) · Dataset samples: StylusAI paper

Three conditions.
One handwriting generator.



StylusAI · §3 and implementation details · Generation schematic; no intermediate model samples shown. Original figure in the appendix.

Handwriting style *across languages*

Target text

fünf

+

Writer style

English writer

learned style embedding

+

Printed character guide

fünf

→

German text
in the selected style

StylusAI



WordStylist · comparison



Stored outputs · German words in IAM English-writer styles · joint IAM + DHSD training

Downstream handwriting recognition

Real DHSD → real + StylusAI data

11.13% → 10.24%

Character error rate ↓

Quality varies by style

StylusAI · Qualitative comparison and Eng-DHSD recognition experiment · CER: 11.13 ± 0.07% / 10.24 ± 0.09%

Since the *first colloquium.*

Previously presented

THE FOUNDATION

DHSD + StylusAI

German handwriting:
dataset + generation

THEN, A FUTURE DIRECTION

Document forgery
detection

Current contributions

SYNTHESIS & UNDERSTANDING

training pairs →

DocDjinn

Generate annotated pages

DocScribe

Recover document structure

DOCUMENT INTEGRITY

DocForgeNet

Localize alterations

JPEG-grid study

Test the detector’s cues

Two connected document research streams · synthesis & understanding, and integrity

From seed documents *to annotated pages*

01 · Select representative seeds

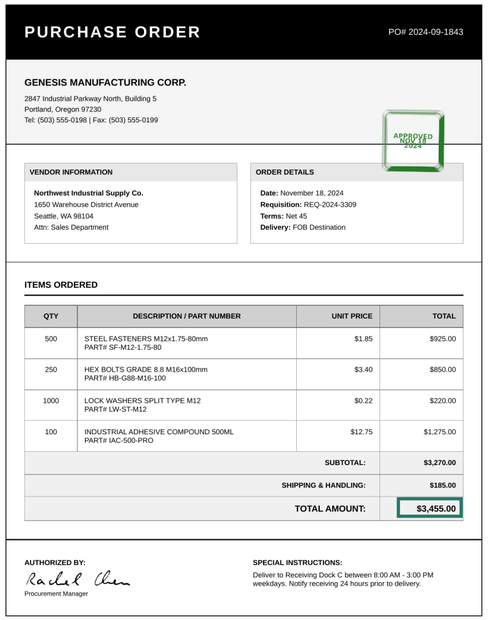


Unlabeled documents · schematic

02 · VLM + task prompt

```
<h1>Purchase order</h1>
<table> … </table>
<div>Total: $3,455.00</div>
```

Illustrative HTML



PURCHASE ORDER

PO# 2024-09-1843

GENESIS MANUFACTURING CORP.
2847 Industrial Parkway North, Building 5
Portland, Oregon 97230
Tel: (503) 555-0198 | Fax: (503) 555-0199

APPROVED
ROUTE
2024

VENDOR INFORMATION

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ORDER DETAILS

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		TOTAL AMOUNT:	**$3,455.00**

AUTHORIZED BY:
Rachel Chen
Procurement Manager

SPECIAL INSTRUCTIONS:
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Rendered DocDjinn sample

03 · Task annotations

What is the total amount?

$3,455.00

Illustrative QA pair

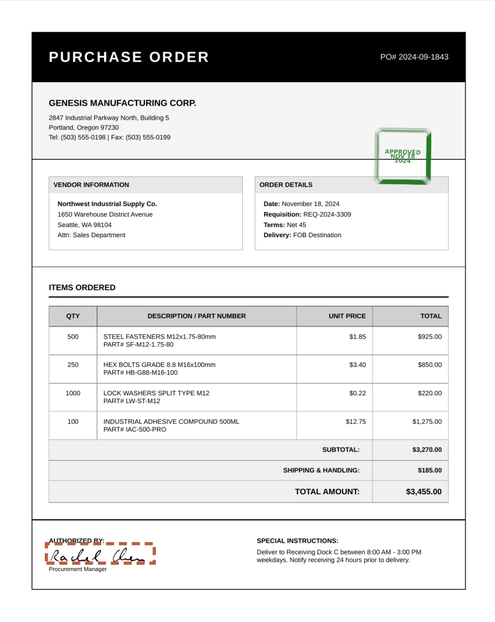
04 · Extract & verify

$3,455.00

Check the annotation against the rendered page

DocDjinn · §3 · Process and annotations are illustrative; the stored page is not an exact seed-to-output trace.

From generated words *to a handwritten line*



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APPROVED NOV 14 2024

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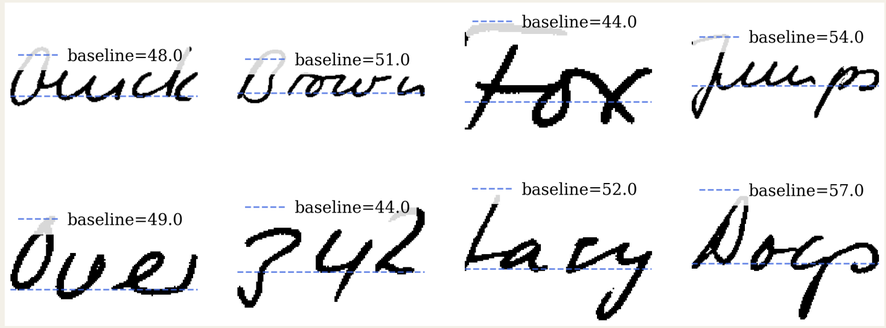
Handwriting in a rendered page

Separate document example

Condition the region

Text + writer style

01 · Generate word segments



02 · Align the baselines

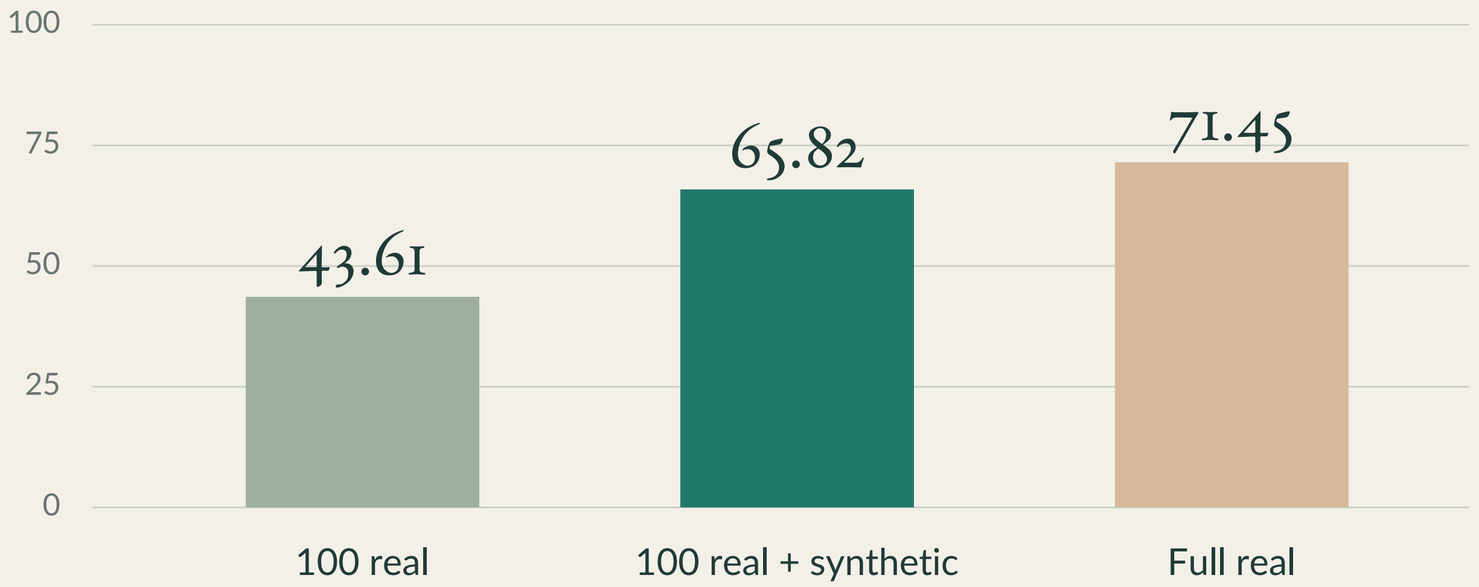


03 · Fit the region; retain its text and spatial annotations

DocDjinn · §3, handwriting synthesis · Original word and sentence outputs; placement connection is schematic.

Synthetic data helps *when labels are scarce*

Answer similarity · ANLS (0–100) ↑



+22.21

points with synthetic data

Starting from

100 real samples

Much of the gap closes.
The full-real reference
remains higher.

One comparison from eleven benchmarks · Gains vary by task and dataset

DocDjinn · Table 1, DocVQA / LayoutLMv3 · Few_B = 100 real samples · Exact ANLS: 43.61 / 65.82 / 71.45

What evidence makes a document *trustworthy?*

First colloquium · future direction

Forgery detection

Current work

Localize alterations.
Test the detector’s cues.

DOCUMENT INTEGRITY

DocForgeNet
Localize alterations

JPEG-grid study
Test the detector’s cues

Keep the content.
Change the compression grid.

JPEG encodes an image in
8 × 8 pixel blocks.

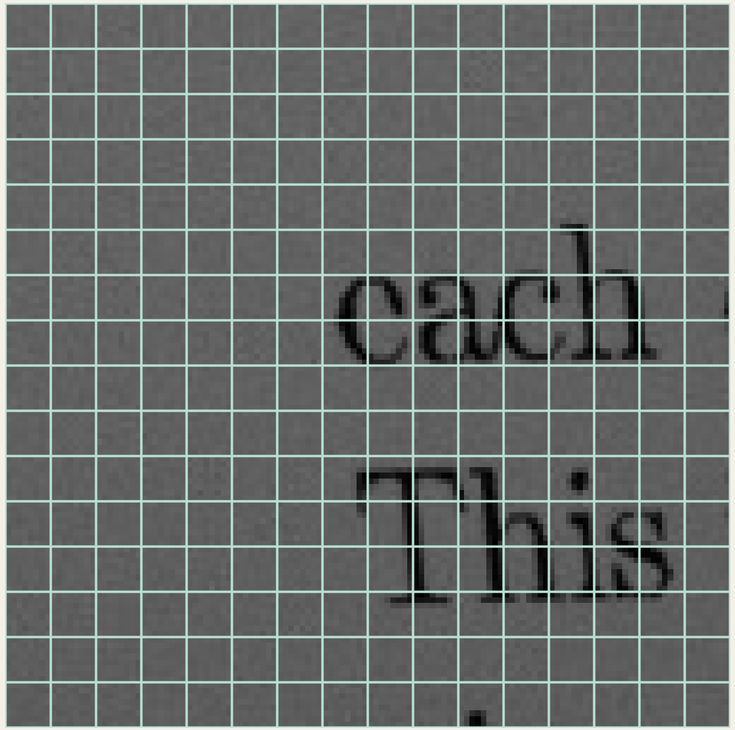
01 **Pad**
4 pixels at the left and top

02 **Recompress**
The same JPEG settings as the baseline

03 **Crop back**
Restore the original size; no recompression after cropping

The document already contains a forgery. PRC introduces no new textual edit.

Original block alignment



128 × 128 px detail · schematic grid

Riaz et al., Mistaking Periodicity for Manipulation, §4-5 · Fixed shift (4, 4); 1-3 JPEG recompressions, quality ≥75

Keep the content.
Change the compression grid.

JPEG encodes an image in
8 × 8 pixel blocks.

01 **Pad**
4 pixels at the left and top

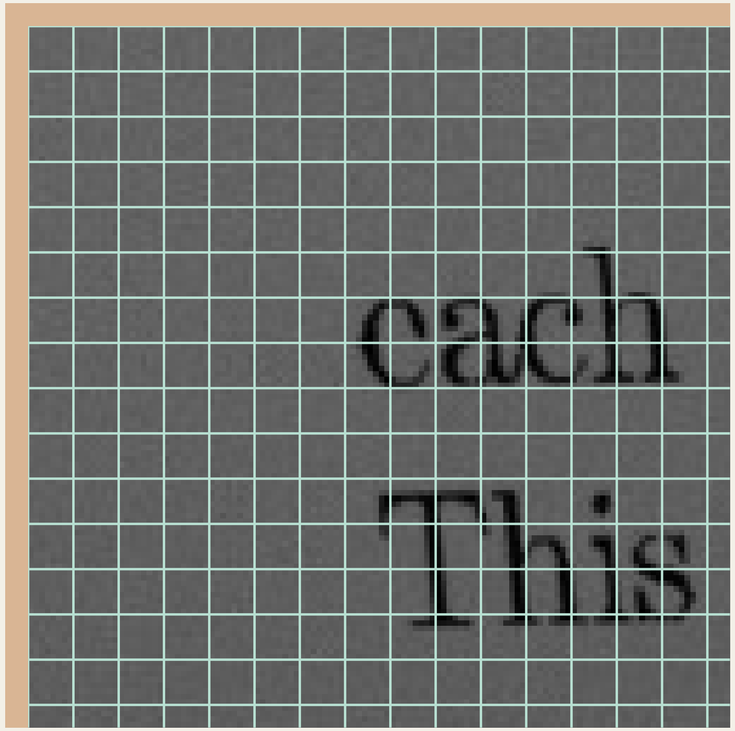
02 **Recompress**
The same JPEG settings as the baseline

03 **Crop back**
Restore the original size; no recompression after cropping

The document already contains a forgery. PRC introduces no new textual edit.

Riaz et al., Mistaking Periodicity for Manipulation, §4-5 · Fixed shift (4, 4); 1-3 JPEG recompressions, quality ≥75

Pad by (4, 4) pixels



128 × 128 px detail · schematic grid

Keep the content.
Change the compression grid.

JPEG encodes an image in
8 × 8 pixel blocks.

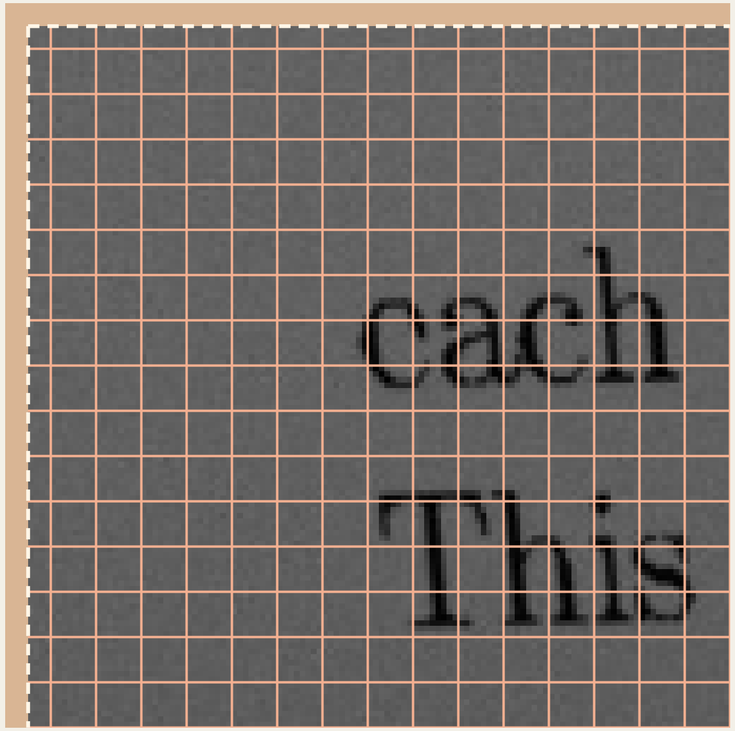
01 **Pad**
4 pixels at the left and top

02 **Recompress**
The same JPEG settings as the baseline

03 **Crop back**
Restore the original size; no recompression after cropping

The document already contains a forgery. PRC introduces no new textual edit.

Recompress on the padded grid



128 × 128 px detail · schematic grid

Riaz et al., Mistaking Periodicity for Manipulation, §4-5 · Fixed shift (4, 4); 1-3 JPEG recompressions, quality ≥75

Keep the content.
Change the compression grid.

JPEG encodes an image in
8 × 8 pixel blocks.

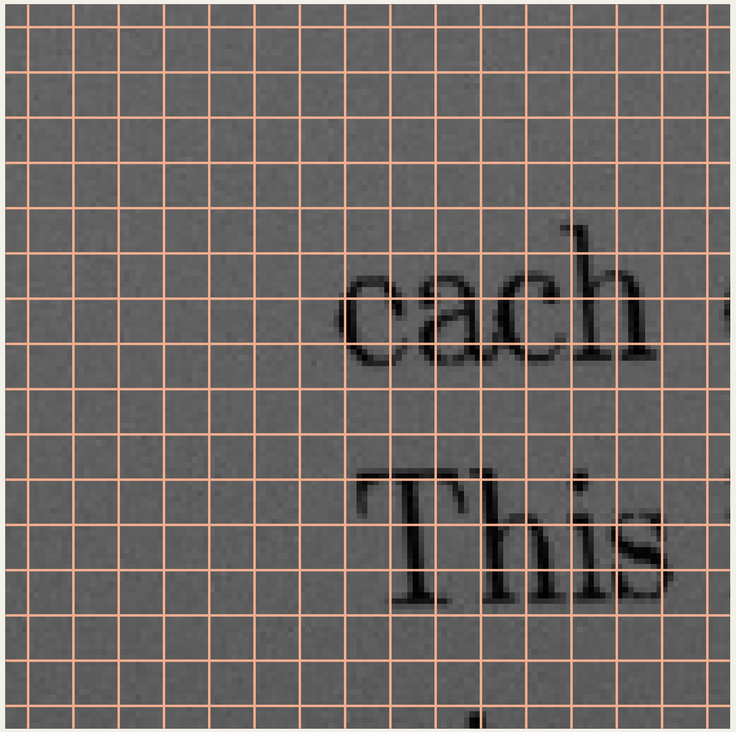
01 **Pad**
4 pixels at the left and top

02 **Recompress**
The same JPEG settings as the baseline

03 **Crop back**
Restore the original size; no recompression after cropping

The document already contains a forgery. PRC introduces no new textual edit.

Crop back: a half-block shift remains



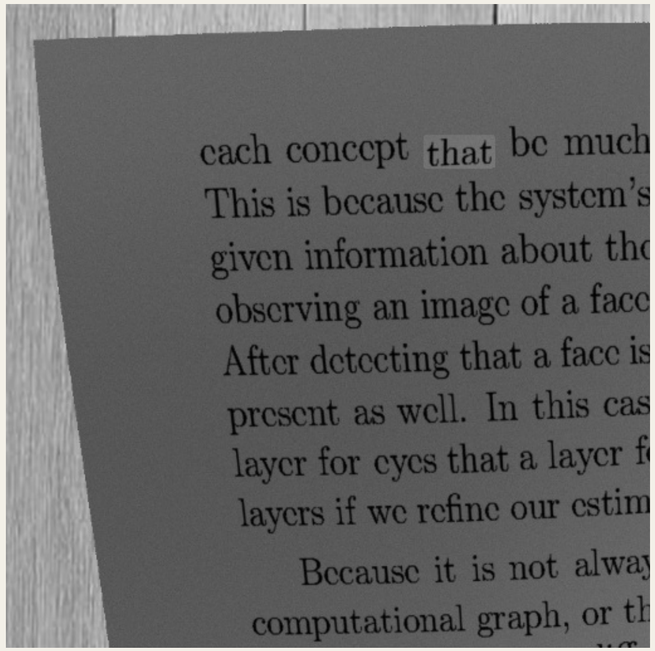
128 × 128 px detail · schematic grid

Riaz et al., Mistaking Periodicity for Manipulation, §4-5 · Fixed shift (4, 4); 1-3 JPEG recompressions, quality ≥75

A small grid shift.
A large change in prediction.

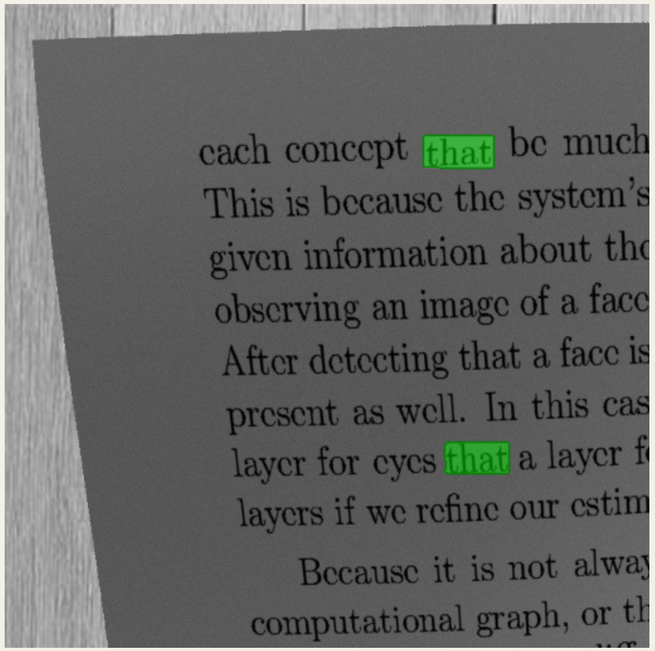
Green: predicted tampering. Same text and layout; recompression can alter pixel values.

01 Input document



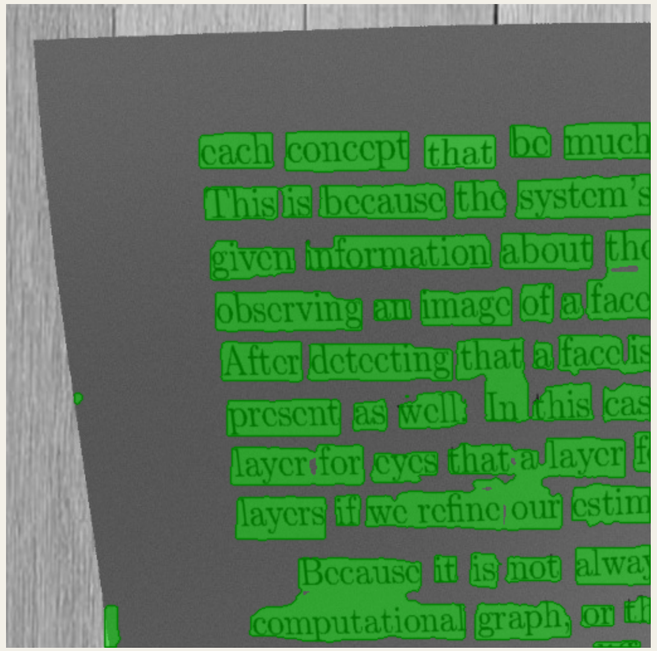
Existing document forgery

02 Baseline prediction



Original compression alignment

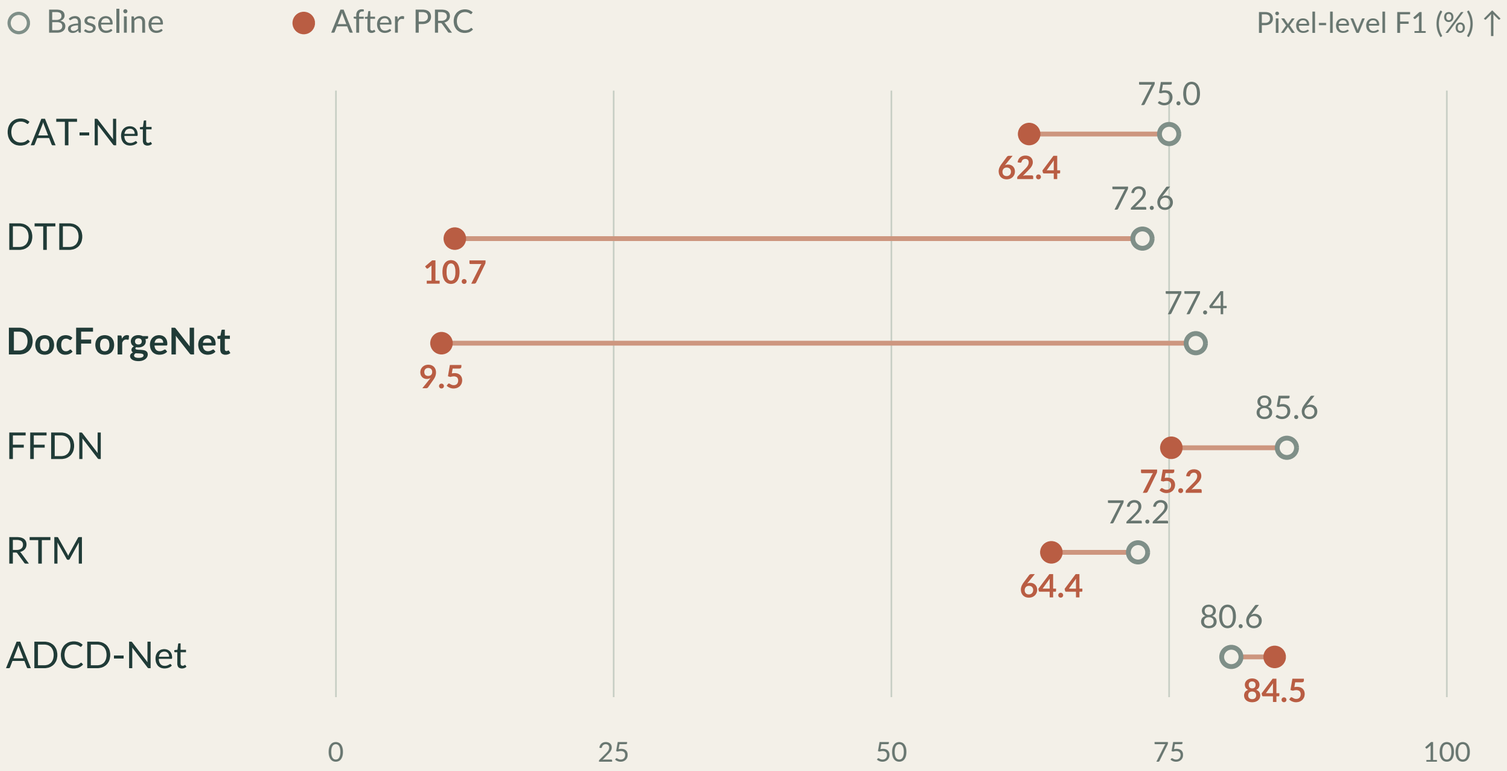
03 After PRC



Extensive additional detections

Riaz et al., Mistaking Periodicity for Manipulation, Fig. 1b · The manuscript does not identify the detector in this example.

The response depends *on the detector*



Riaz et al., Mistaking Periodicity for Manipulation, Table 1 · FPA = false-positive pixels / total image area, averaged over images

DocForgeNet

77·4 → 9·5

F1 (%) after the grid shift

21.5%

of the total image area falsely marked, on average

ADCD-Net improves in this condition. The effect is not universal.

Compression structure *can become a fragile cue.*

What this supports 01

Changing JPEG-grid alignment can substantially change localization, even when the document's text and layout are preserved.

What this leaves open 02

Recompression also changes some RGB values. The experiment does not isolate every image cue, and responses vary across models.

Scope: evaluated JPEG-based RGB+DCT detectors · fixed half-block shift

Riaz et al., Mistaking Periodicity for Manipulation, Quantitative Evaluation and Limitations

From handwriting *to document reliability.*

A broader research programme. A more demanding question.

Previously presented

THE FOUNDATION

DHSD + StylusAI

German handwriting:
dataset + generation

THEN, A FUTURE DIRECTION

Document forgery detection

Current contributions

SYNTHESIS & UNDERSTANDING

training pairs →

DocDjinn

Generate annotated pages

DocScribe

Recover document structure

DOCUMENT INTEGRITY

DocForgeNet

Localize alterations

JPEG-grid study

Test the detector's cues

Useful data · Faithful structure · Tested evidence

Thank you.

Questions & discussion

Useful data.
Faithful structure.
Tested evidence.

Nauman Riaz

RPTU Kaiserslautern-Landau · DFKI

Reliability must be tested beyond familiar conditions.

RETURN TO THE EVIDENCE

Select a number or title to revisit a slide

Starting point

01 Opening
02 DHSD recap
03 StylusAI model
04 StylusAI evidence

Synthesis & understanding

05 Research map
06 DocDjinn pipeline
07 Handwriting
08 DocVQA results

Document integrity

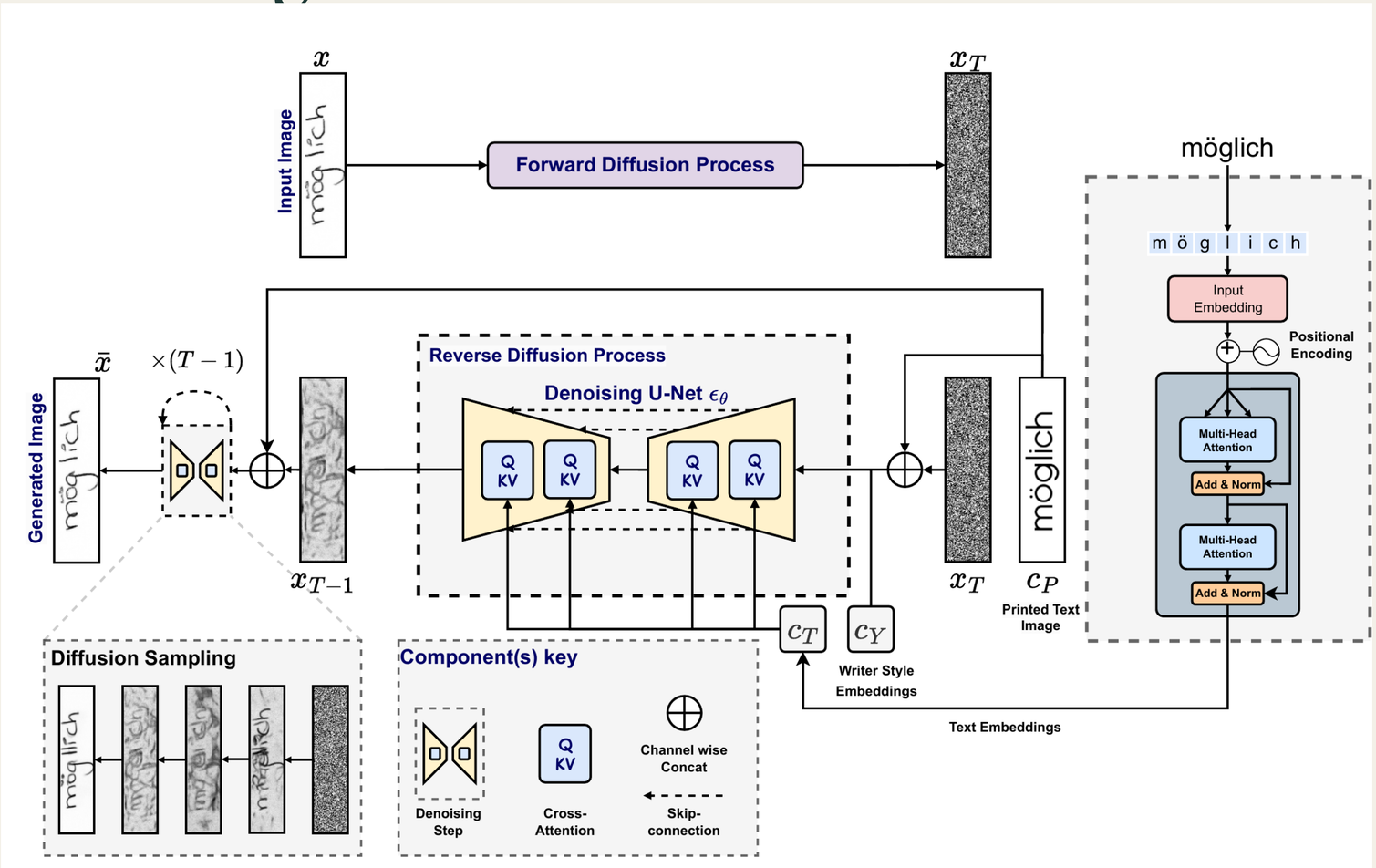
09 Integrity
10 PRC mechanism
11 Predictions
12 Detector results
13 Limits & meaning

Conclusion

14 Research synthesis
15 Closing index

Original model diagram

Return to architecture →



StylusAI · Original architecture figure, unchanged. The method prose additionally specifies writer + timestep embedding addition; the main slide makes this explicit.